Farmer Crop Insurance Claim Tracking System – Bihar

Proposed Solution

To solve the delays, lack of transparency, and paperwork issues in Bihar's crop insurance claim process, we propose a **digital Crop Insurance Claim Tracking System** that provides a centralized platform for farmers, officers, and administrators.

This system will ensure that farmers can submit claims easily, upload documents online, track claim status in real-time, and receive notifications without repeatedly visiting government offices.

1. Farmer Claim Registration Portal
2. Digital Document Upload & Verification
3. Claim Workflow and Status Tracking System
4. Officer Verification Dashboard
5. Notification & Reminder System
6. Secure Role-Based Access System

System Architecture Overview

Frontend

- Farmer portal (claim submission, document upload, status tracking)
- Officer portal (verification, approval/rejection updates)
- Admin portal (dashboard, reports monitoring)
- Mobile-friendly web interface

Backend

- Handles login and authentication
- Processes claim registration and claim ID generation
- Validates uploaded documents
- Manages claim workflow and status updates
- Supports officer verification and approval
- Sends notifications and reminders

- Generates reports and dashboards

Database / Storage

- Stores farmer details and records
- Stores claim details and disaster type information
- Stores claim status and approval history
- Stores claim logs and audit trail
- Stores uploaded documents (PDF/JPG/PNG) in file storage

Key Components of the Proposed System

- Farmer Registration Module
- Claim Submission Module
- Document Upload & Management Module
- Claim Status Tracking Module
- Officer Verification Module
- Approval / Rejection Module
- Notification & Reminder Module
- Dashboard Module (Farmer/Officer/Admin)
- Report Generation Module
- Claim Logs & Approval History Module
- Disaster Type Management Module
- Database & Record Storage Module
- Authentication & Role Management Module

Data Flow Explanation (Simple)

1. **Farmer Registration**
    - Farmer enters details (name, mobile, Aadhaar, address).
    - Data is saved in the **Farmer Database**.

2. **Claim Submission**
    - Farmer fills claim form (crop type, disaster type, loss details).
    - System generates a **unique Claim ID**.
    - Claim details are stored in the **Claim Database**.
3. **Document Upload**
    - Farmer uploads required documents (Aadhaar, land proof, bank details, photos).
    - Documents are stored in **File Storage** and linked to the claim.
4. **Claim Processing**
    - Backend checks if documents are complete.
    - If missing, claim status becomes **Pending Documents**.
5. **Officer Verification**
    - Officer views claims through officer dashboard.
    - Officer verifies documents and updates claim status.
6. **Status Update & Logging**
    - Every action (verification, approval, rejection) is saved in the **Claim Logs Table**.
    - Approval history is updated.
7. **Notifications**
    - When status changes, farmer gets an alert (SMS/email/app notification).
8. **Final Decision**
    - If approved → claim moves to **Payment Processing**.
    - If rejected → reason is stored and farmer is notified.
9. **Admin Monitoring**

- Admin views dashboards and generates reports using claim data from database.